# Unit 4: Energy, Environment, Microbiology and Immunity

## Topic 5 – Energy Flow, Ecosystems and the Environment

**Candidates will be assessed on their ability to:**

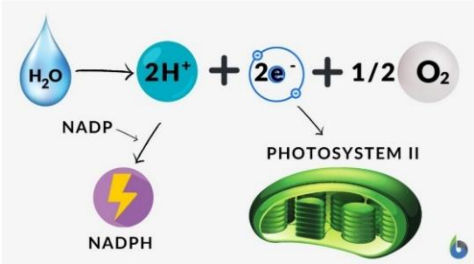
**5.1** understand the overall reaction of photosynthesis as requiring energy from light to split apart the strong bonds in water molecules, storing the hydrogen in a fuel (glucose) by combining it with carbon dioxide and releasing oxygen into the atmosphere

**Overall reaction of photosynthesis:**

$$6\,CO_2 + 6H_2O \xrightarrow[\text{CHLOROPHYLL}]{\text{LIGHT}} C_6H_{12}O_6 + 6\,O_2$$

**Photolysis of water:**

- Occurs in thylakoid membranes of chloroplasts
- Covalent bond of a water molecule is broken
- Supply **hydrogen ions** to reduce carbon dioxide which is then stored as glucose
- Supply **electrons** for photosystem II (680nm)
- **Oxygen** is produced as a waste product

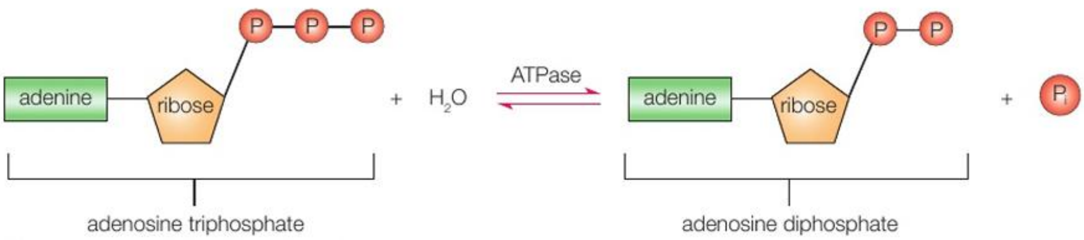



**Light Energy:**

- Light energy cannot be used directly as ATP is the source of energy

1. Light energy is absorbed by photosystems
2. This excites electrons
3. These electrons are passed along a series of electron carriers
4. Therefore, **releasing energy (NOT ATP)** to phosphorylate ADP into ATP (cyclic)]
5. Phosphorylation of ADP via the proton gradient to form ATP (non-cyclic) →This is because it also produces hydrogen ions which can be used in chemiosmosis

**5.2** understand how photophosphorylation of ADP requires energy and that hydrolysis of ATP provides an immediate supply of energy for biological processes




**Forward Reaction:**

- Hydrolysis
- **Releases** energy (34kJ)

**Backward Reaction:**

- Condensation
- **Needs** energy (34kJ)

*Both require **ATPase** enzyme as catalyst.

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| **5.3** | understand the light-dependent reactions of photosynthesis, including how light energy is trapped by exciting electrons in chlorophyll and the role of these electrons in generating ATP, reducing NADP in cyclic and non-cyclic photophosphorylation and producing oxygen through photolysis of water |

**Light-Dependent Reaction:**

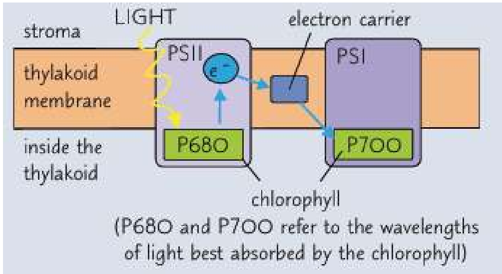
→This occurs in thylakoid membrane
→ATP is made by photophosphorylation in light-dependent reaction.

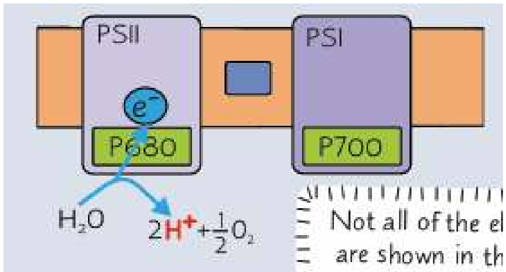
**(2) types of Photophosphorylation**

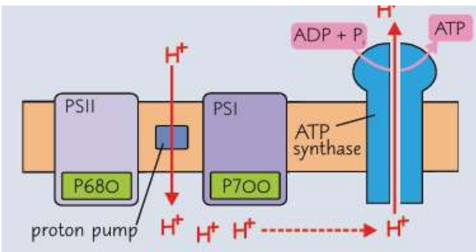
1. **Non-Cyclic Photophosphorylation**
2. **Cyclic Photophosphorylation**

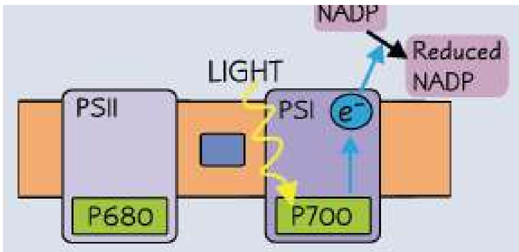
**1.Non-Cyclic Photophosphorylation**

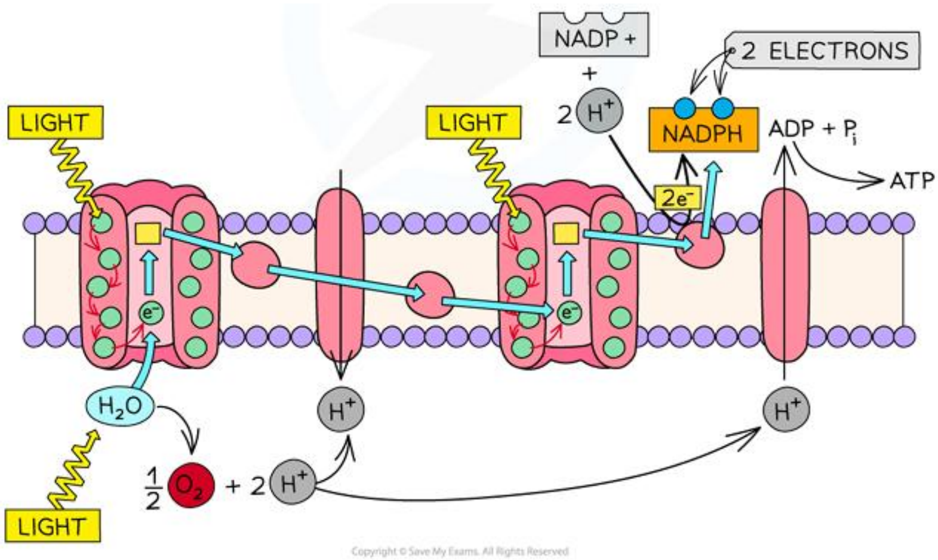
1.PSII absorbs light energy to **excite** electron to move to higher energy level.
2.The electron in PSII is replaced by electron from **photolysis** of water.
3. The excited electron loses energy as it moves along the **electron transport chain**.
→ This **energy** (NOT ATP) transports protons into thylakoid intermembrane space to form proton gradient and this produces **ATP** (chemiosmosis).
4. Light energy is absorbed by PSI, which excites the electron again to even higher energy level.
→ The electron is finally transferred to NADP along with hydrogen ion from the stroma to form **reduced NADP**.



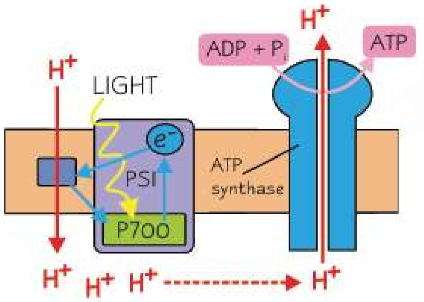



**2.Cyclic Photophosphorylation**

→Only uses PSI.
→It is 'cyclic' as the electrons from chlorophyll molecule are not passed onto NADP, but are passed back to PSI via electron carrier.
→This process does not produce any reduced NADP or $O_2$, only produces small amounts of ATP.




***Both reduced NADP and ATP enter the light-independent reaction:**
→ ATP is required to provide energy to convert GP to GALP
→ Reduced NADP is used to reduce GP to GALP

**5.4**

(i) understand the light-independent reactions as reduction of carbon dioxide using the products of the light-dependent reactions (carbon fixation in the Calvin cycle, the role of GP, GALP, RuBP and RUBISCO)

(ii) know that the products are simple sugars that are used by plants, animals and other organisms in respiration and the synthesis of new biological molecules (polysaccharides, amino acids, proteins, lipids and nucleic acids)

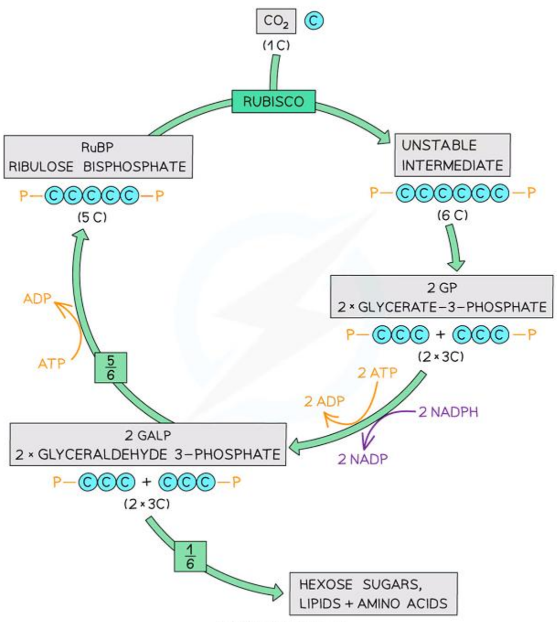
**i**

**Light-Independent Reaction (Calvin Cycle):**

→ Calvin cycle happens inside the stroma of chloroplast.

**(4) Steps of Calvin cycle:**

1. **Carbon Fixation:** RUBP (Ribulose Biphosphate) (5C compound) fixes $CO_2$ to produce (2) GP (Glycerate-3-Phosphate) (3C compound).
   →This is catalyzed by **RUBISCO** (Ribulose Biphosphate Carboxylase).
2. **Reduction:** GP is reduced into GALP (Glyceraldehyde 3-Phosphate) using hydrogen from **reduced NADP** and energy from hydrolysis of **ATP**.
3. **Regeneration:** GALP is converted back into RUBP.
   → This uses energy from hydrolysis of **ATP**.
4. **Glucose Formation:** Every **6th turn of the Calvin cycle**, (1) molecule of glucose is produced from GALP.




**ii**

**Products of Light Independent Reactions:**

- GALP converted glucose every 6th turn of Calvin cycle

**Cellulose:**

- Beta-glucose arranged in reverse structure to form 1,4-glycosidic bonds

**Lipid:**

- Glucose is used to make glycerol and fatty acids

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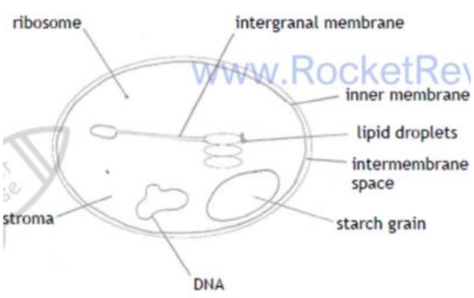
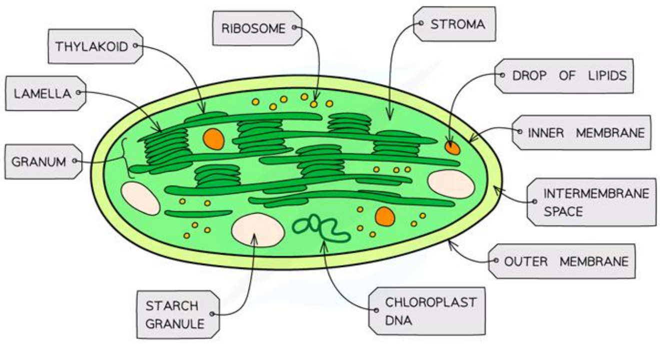
| | - Glycerol and fatty acids joined by ester bonds in a condensation reaction, forming lipids<br>**DNA:**<br>- Glucose is converted into deoxyribose<br>- Then, nitrogen from nitrates is added to glucose, forming a nitrogenous base and phosphate from phosphate ions are added to form DNA nucleotide<br>**Proteins:**<br>- Amino acids made from glucose and nitrates taken up from the soil<br>- Amino acids joined by peptide bonds<br>- Glucose respired to produce ATP for protein synthesis<br>**Biomass:**<br>- Glucose and fructose are used to produce sucrose<br>- Amino acids made from glucose and nitrates taken up from the soil<br>- Sucrose and amino acids are transported to the roots through the phloem<br>- They are used to synthesize organic molecules such as starch, cellulose and proteins<br>- Calcium ions are used to make calcium pectate for cell walls |
| **5.5** | understand the structure of chloroplasts in relation to their role in photosynthesis |
| | <br><br>(ii) Compare and contrast the structure of the outer membrane of a chloroplast with that of a thylakoid membrane.<br><br>(4)<br>Both consist of phospholipid bilayer<br>Both have proteins<br>Thylakoid membrane has ATP synthase but outer membrane does not<br>Thylakoid membrane has chlorophyll / PSI / PSII while outer membrane does not<br>Thylakoid membrane has electron carriers but outer membrane does not |

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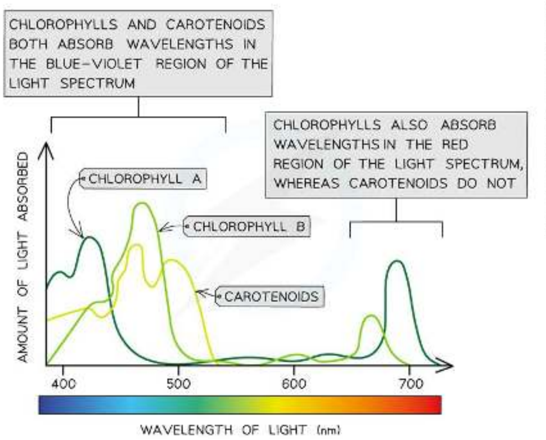
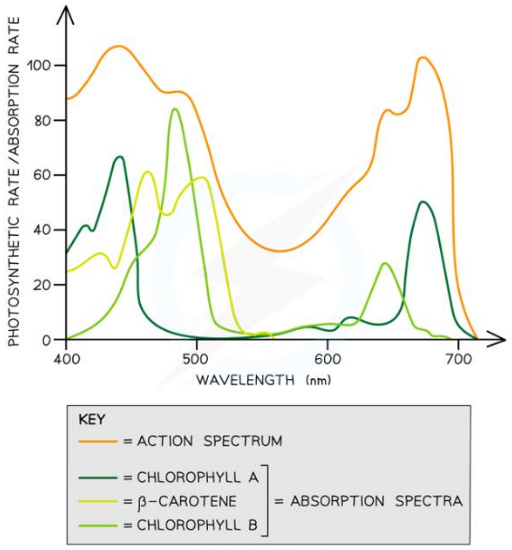
| | **Structures with Functions:**<br>- Chloroplast envelope: space between inner and outer membrane<br>- Grana: stacks of thylakoids<br>- **Thylakoids:**<br>→ double membrane; contains photosystems<br>→ **light dependent reaction** occurs in **thylakoid membrane**<br>→ this is where chlorophyll captures light and uses it to produce ATP<br>- Lamellae: connects grana<br>- **Stroma:**<br>→ gel-like fluid that contains all the enzymes needed for photosynthesis<br>→ **light independent reaction** occurs (Calvin cycle)<br>- DNA<br>- Ribosomes |
| **5.6** | understand what is meant by the terms *absorption spectrum* and *action spectrum* |
| | **Absorption Spectrum:** Rate of absorption of light at different wavelengths by different pigments<br><br>**Action Spectra:** Rate of photosynthesis with wavelength of light<br><br>*To determine color of the pigment based on the wavelength of light absorbed:<br>- Chlorophyll A: Do not absorb yellow-green so its color is yellow-green<br>• So, the color of wavelength it does not absorb is the color of pigment because that unabsorbed light is reflected back to give its color<br><br><br><br><br> |
| **5.7** | understand that chloroplast pigments can be separated using chromatography and the pigments identified using Rf values |

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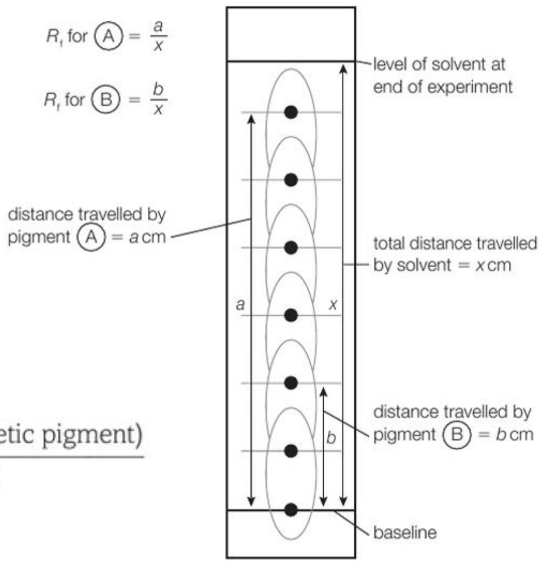
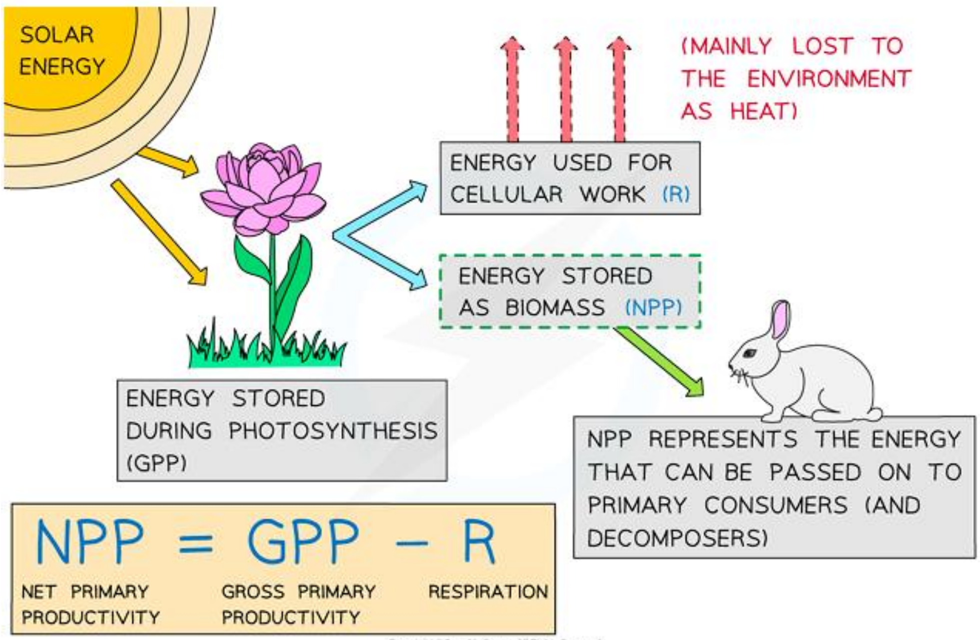
| | **How a chromatogram can be produced:**<br>- Pigment dissolved in a solvent (ethanol/water)<br>- Pigment will then be loaded onto the filter paper<br>- Filter paper will then be dipped into the same solvent<br>- This is left to run until the leading edge is near the end of filter paper<br><br>**To determine $R_f$ value:**<br>$R_f \text{ value} = \frac{\text{distance travelled by solute (photosynthetic pigment)}}{\text{distance travelled by solvent}}$<br> |
| **5.8** | **CORE PRACTICAL 10**<br>**Investigate the effects of light intensity, light wavelength, temperature and availability of carbon dioxide on the rate of photosynthesis using a suitable aquatic plant.** |
| **5.9** | (i) understand the relationship between gross primary productivity (GPP), net primary productivity (NPP) and plant respiration (R)<br>(ii) be able to calculate net primary productivity |
| **i** | **Gross Primary Productivity:** the rate at which light energy is converted into carbohydrates during photosynthesis<br>**Net Primary Productivity:** the rate at which light energy is stored in plant biomass<br>**R (Respiratory Losses):** the carbohydrates used up by the plant that are not stored in plant biomass<br>→**Units for ALL:** kJ $m^{-2}$ $yr^{-1}$ |
| **ii** |  |
| **5.10** | know how to calculate the efficiency of biomass and energy transfers between trophic levels |

**Between Sun and Producer (Plant):**

- Not all wavelengths of light can be absorbed by the plant
- Green light is reflected off the leaf
- Not all the light falls on the leaves
- Passes through leaf

**Between Consumers:**

- Inedible parts in organisms
- Energy lost in respiration as heat
- Egestion of indigestible parts of organism
- Decomposers feed on the dead organisms by aerobic respiration

**Why trophic level is limited:**

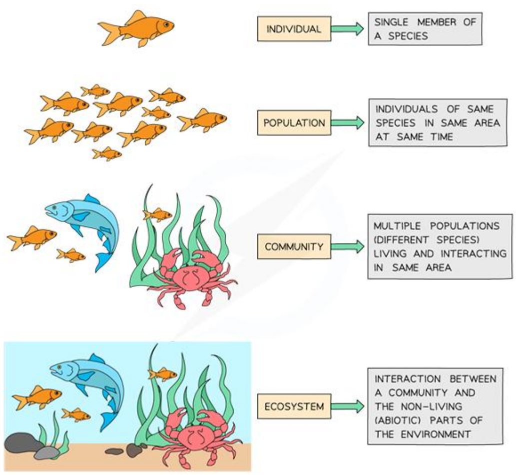
- Energy is lost between trophic levels
- So not enough energy for another trophic level

$$\boldsymbol{Percentage\ Difference = \frac{(Final\ Value - Initial\ Value)}{Initial\ Value} \times 100}$$

$$\boldsymbol{Mean\ Percentage\ Difference\ between\ A\ and\ B\ = \frac{A - B}{\frac{A + B}{2}} \times 100}$$

$$\boldsymbol{Percentage\ Efficiency\ of\ Energy\ Transfer = \frac{Energy\ available\ in\ one\ trophic\ level}{Energy\ available\ in\ previous\ trophic\ level} \times 100}$$

**5.11** understand what is meant by the terms *population*, *community*, *habitat* and *ecosystem*




**Population:** organisms of one species found in a particular area

**Community:** a group of organisms of different species interacting in a particular area

**Habitat:** a place where organisms live

**Ecosystem:** all organisms interacting with each other and the abiotic factors

**Biodiversity:** variety of different species in a habitat and the genetic variation (different alleles) within those species

**Biomass:** how much organic matter present in organisms

**5.12** understand that the numbers and distribution of organisms in a habitat are controlled by biotic and abiotic factors

**Abiotic Factors:**

**1.Light:** → Influence photosynthesis rate → Affect plant growth rate

**2.Temperature:** → Affect enzyme activity in animals and plants (RUBISCO)
→ Higher temperature results in higher kinetic energy of substrates so form more enzyme-substrate complexes

**3.Water Availability:** → For photosynthesis (light-dependent reaction) and for drinking for animals

**4.Oxygen Availability:** → For aerobic respiration;

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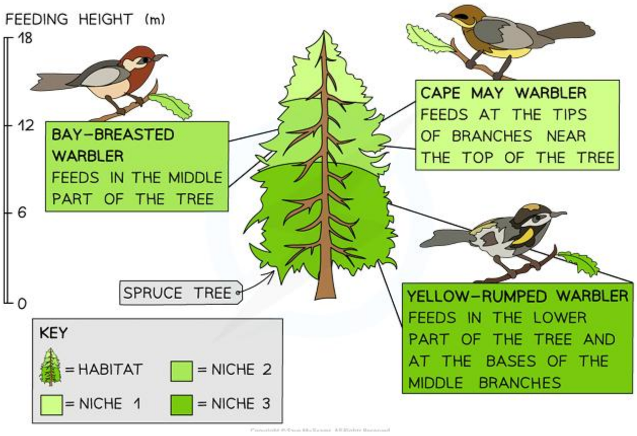
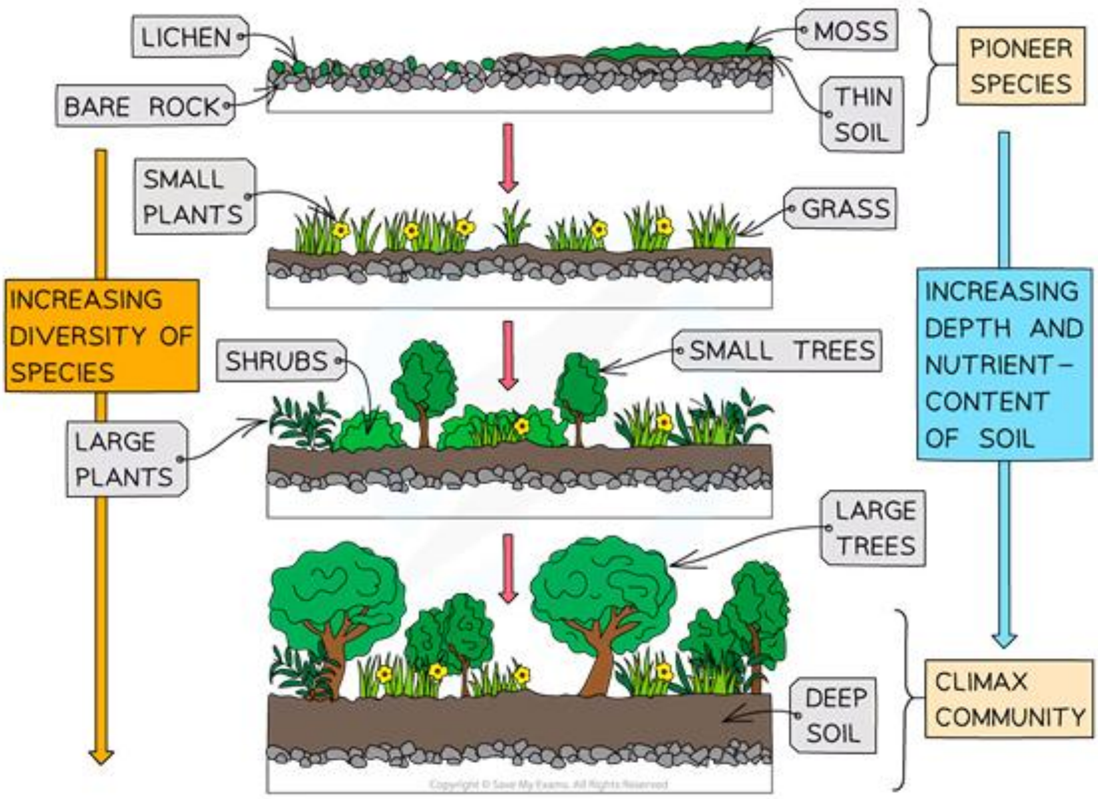
| | → In soil, active transport process in roots for uptake of minerals<br>→ In ocean, for aquatic animals<br><br>**5.Edaphic Factors (Soil Structure and Mineral Content)**<br>→ Loam is the ideal soil as leaching of minerals is less likely and easy to drain water<br><br>**Biotic Factors:**<br><br>**1.Predation:** → Population of both predator and prey will oscillate in a repeating cycle if other factors (such as parasitism) are not affected<br><br>**2.Territory:** → Purpose is to ensure that breeding pair has sufficient resources to raise their offspring safely<br><br>**3.Parasitism and Disease:** → If a species is sick, other competing species will outgrow if they are resistant<br>→ The whole population can be wiped out by the parasitism itself or due to the competition by other species with resistant genes<br><br>**4.Finding a Mate:** → Level of breeding depends on available mate founds.<br>→ If mates of same species are not found, hybridization may occur. |
| **5.13** | understand how the concept of niche accounts for the distribution and abundance of organisms in a habitat |
| | **Niche:** role of a living organism in its habitat<br><br><br><br><br>**Adaptation to niche:**<br>- **Anatomical**: adaptations involving the form and structure of an organism<br>- **Physiological**: adaptations involving the way the body of the organism works, including differences in biochemical pathways or enzymes<br>- **Behavioral**: adaptation involving programmed or instinctive behavior making organisms better adapted for survival |
| **5.14** | **CORE PRACTICAL 11**<br><br>**Carry out a study of the ecology of a habitat, such as using quadrats and transects to determine the distribution and abundance of organisms, and measuring abiotic factors appropriate to the habitat.** |
| **5.15** | understand the stages of succession from colonization to the formation of a climax community |

**Succession:** a sequence of changes in a community over a period of time
**Climax Community:** final stage of community that is self-sustaining and stable




**Mechanism of Succession:**

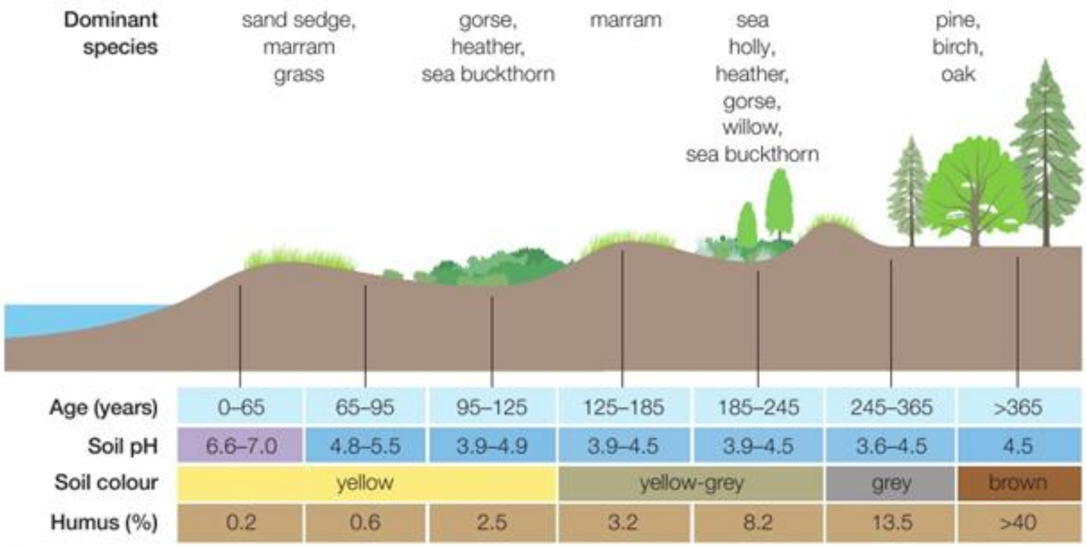
1. **Colonization**: **Pioneer species** such as algae and mosses produce enzymes and acids that digest the surface of bare rocks, releasing organic matter and forming a basic soil layer called **humus**
2. This allows growth for short plants such as grass and shrubs, which forms more humus when they die and decay, which adds the **soil depth**.
   → This improves soil conditions such as soil texture, soil moisture, water and mineral availability.
3. The process continues with different plant species for over years until **climax community** is reached.

**Characteristics of Pioneer Species:**

- Need to withstand harsh environments
- Need to be able to withstand lack of shade
- Have a low requirement for minerals
- Have fast life-cycles and wide dispersal mechanisms

**Why Succession occurs in Stages:**

- Changes to the habitat have to take place before a different organism can survive there
- To make soil improvements
- This allows deeper soil bushes to grow

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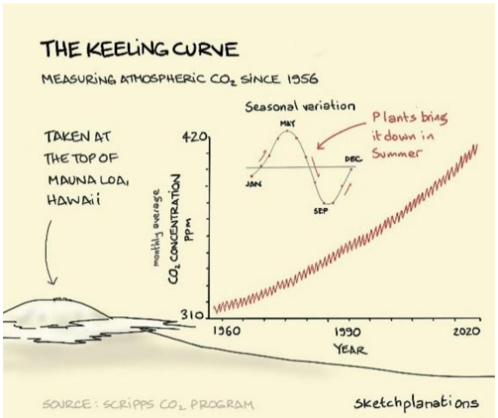
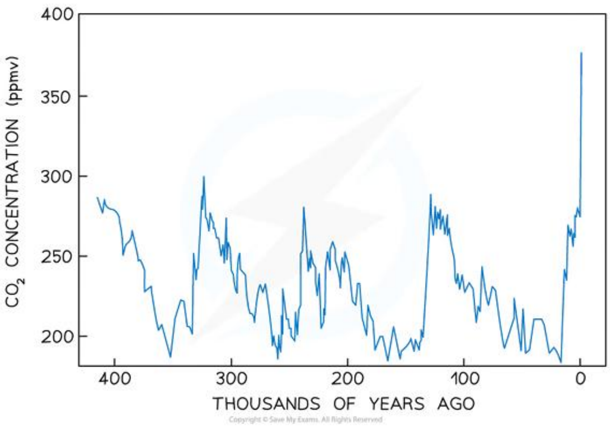
| Age (years) | 0–65 | 65–95 | 95–125 | 125–185 | 185–245 | 245–365 | >365 |
| Soil pH | 6.6–7.0 | 4.8–5.5 | 3.9–4.9 | 3.9–4.5 | 3.9–4.5 | 3.6–4.5 | 4.5 |
| Soil colour | yellow | | | yellow-grey | | grey | brown |
| Humus (%) | 0.2 | 0.6 | 2.5 | 3.2 | 8.2 | 13.5 | >40 |

**Changes in Soil Structure:**
→**Soil depth:** Increases with distance from sea as there are older sand dunes so more time for humus to build
→**Organic material:** increases with distance from sea as more decomposition adds organic material to the sand
→**pH:** decreases with distance from sea because humus is acidic and more humus is added over time
→**Percentage of bare rock:** decreases because pioneer species break it down
→**Number of different species of plant:** Increases with distance from sea because soil becomes more fertile and salinity is decreasing (basicity) and organic material holds more moisture than sand alone

**5.16** understand the different types of evidence for climate change and its causes, including records of carbon dioxide levels, temperature records, pollen in peat bogs and dendrochronology, recognizing correlations and causal relationships

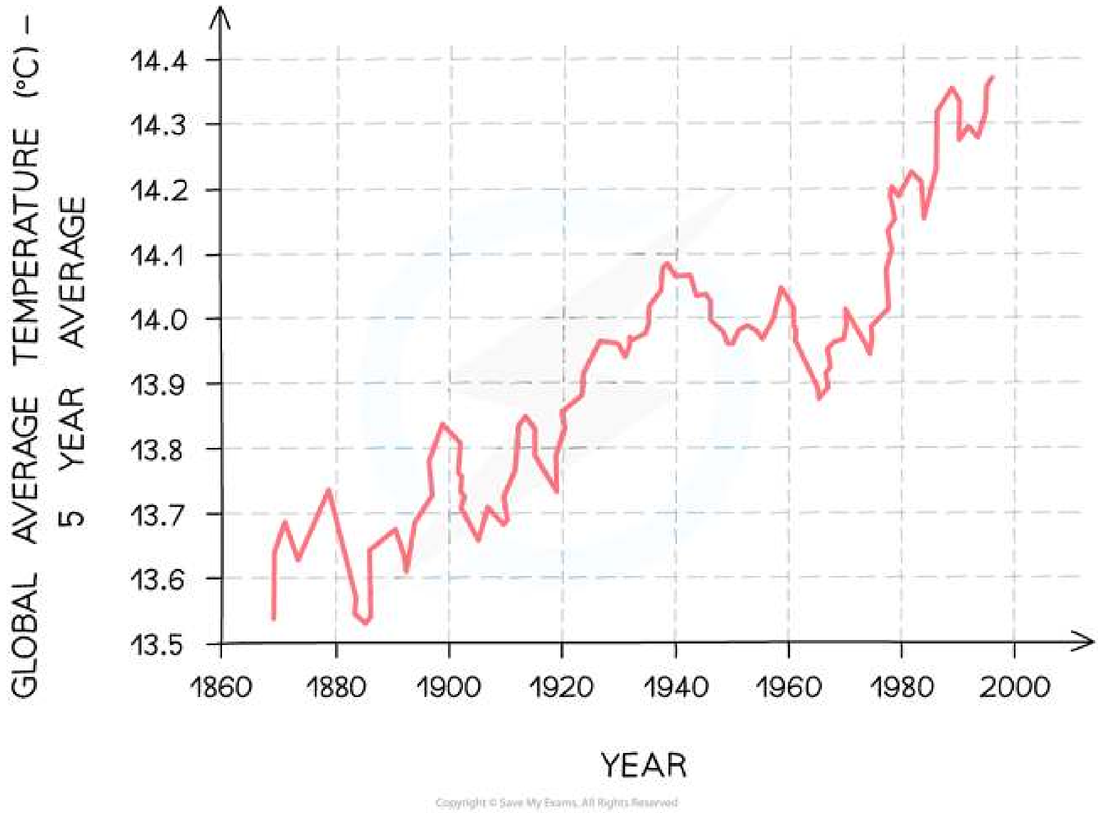
**Evidence for Climate Change:**



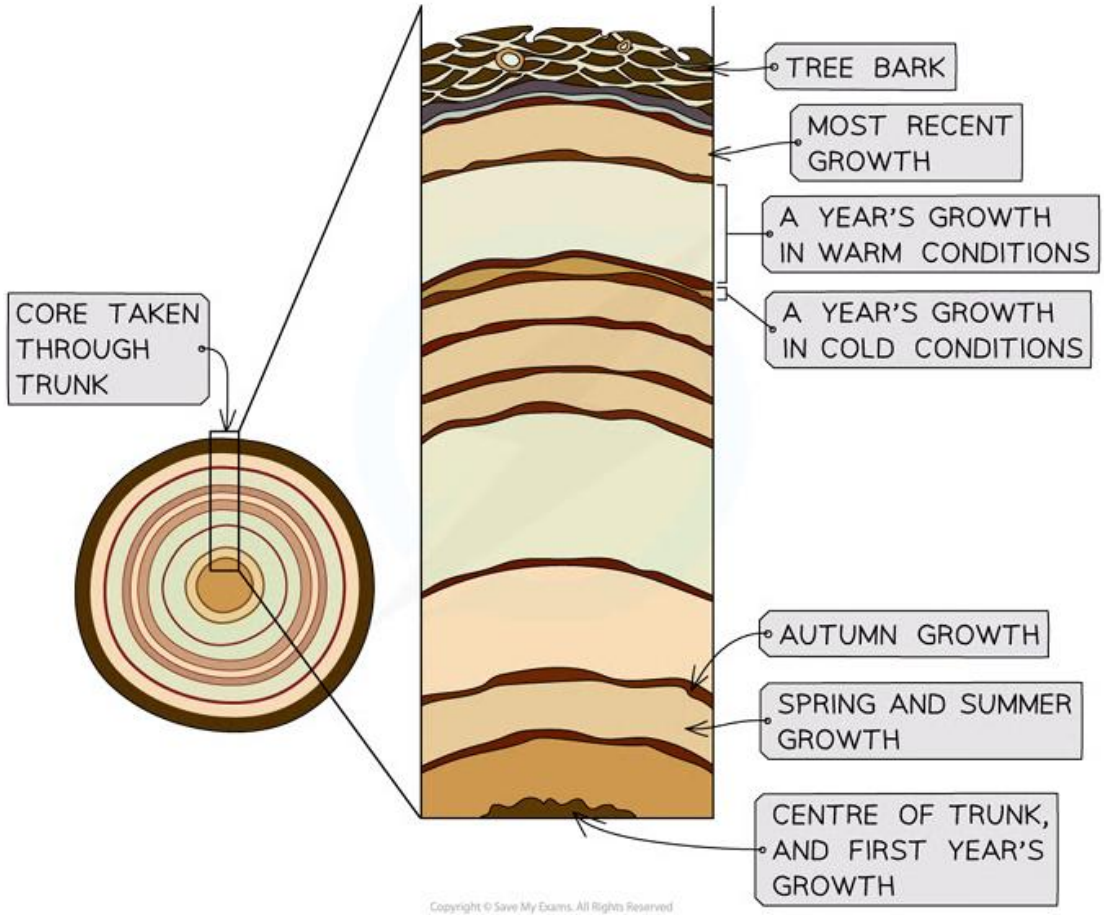

**1.Records of Carbon Dioxide Levels**

- Volcanic eruptions and weathering of limestone rocks contributed mostly in ancient time to the $CO_2$ level in atmosphere
- → This is known from gas composition of bubbles formed in ancient ice cores
- However, since the industrial revolution, atmospheric $CO_2$ levels have skyrocketed to the highest in Earth's history
- Data show a correlation between changing atmospheric $CO_2$ and temperature over thousands of years
- → However, correlation does not equal to causation, but anthropogenic carbon emission has been a strong evidence leading to global warming




**2.Temperature Records**

- Since 1850, temperature has been measured around the world using thermometers
- This gives reliable but short-term record of global temperature change
- Records from that age show overall trend in increasing average global temperatures
- Causes: Anthropogenic $CO_2$ Emission, Solar Flares, Volcanic Eruptions



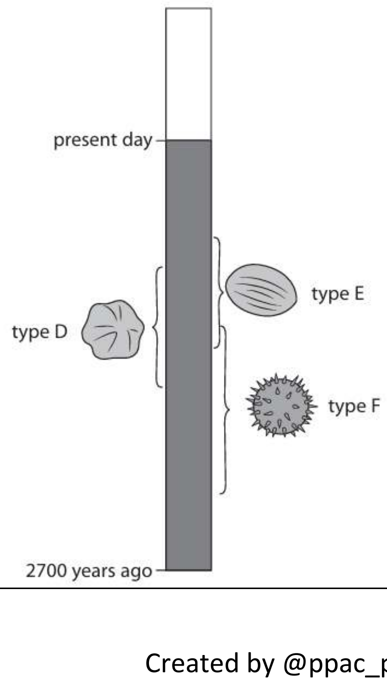

**3.Dendrochronology**

- Dendrochronology is the study of tree rings
- Every year a tree ring (xylem) is formed
- By counting the number of xylem rings, plant's age can be known
- Thickness of the ring is proportional to the temperature of the year in which the ring was formed
- As temperature increases → RUBISCO activity increases → Higher rate of photosynthesis → More GALP is produced → More biomass produced
- However, other factors such as light intensity, rainfall and $CO_2$ concentration can also contribute to thickness so it is not very accurate

**4.Peat Bogs**

- Conditions in bog were not suitable for decomposition to take place
- **Cold temperatures** mean that microbial enzymes were working slowly
- **Low pH** (acidic conditions) is not suitable for bacteria to survive
- Water-logged bog meant that **oxygen levels** were very **low** (conditions are anaerobic)
- So not enough energy released by respiration
- Bog protected from insects and animals

→ **Explain distribution of types of pollen grains in the peat bog / mud core:**




- Each type of pollen grain of one species is present when conditions are favorable for that plant

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- Condition changes overtime, allowing different types of plants to grow<br>
- Previous type of plant disappears due to competition with another species or favorable conditions changed<br>
- Plants compete for light, space, water and nutrients
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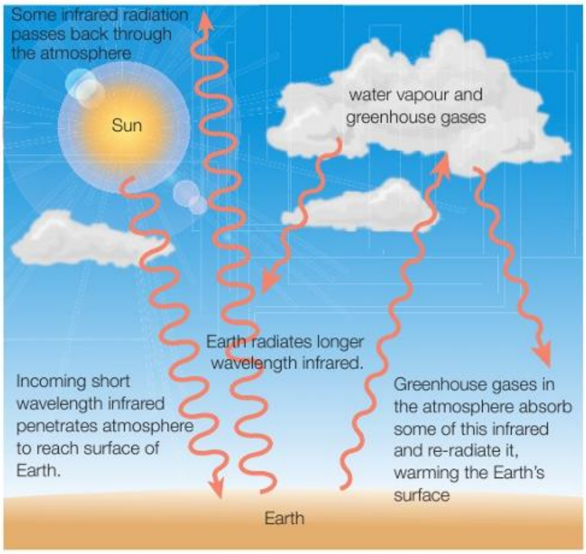
<tr><td>5.17</td><td>understand the causes of anthropogenic climate change, including the role of greenhouse gases in the greenhouse effect</td></tr>
<tr><td></td><td>
Anthropogenic Climate Change: change in mean temperature of an area over a long period of time caused by human activities<br><br>
Greenhouse Gases:<br>
- Carbon dioxide<br>
→ Combustion, Aerobic respiration, Decomposition<br>
- Methane<br>
→ Anaerobic respiration of bacteria in water logged paddy field, Stomach of ruminant animals<br>
→ Decomposition in landfill sites<br>
- Water Vapor<br>
→ Evaporation<br>
- CFC<br>
→ Refrigerator<br><br>
Greenhouse Effect:<br>
1. Incoming short wavelength passes through the greenhouse gases and reach surface of the Earth<br>
2. The Earth surface radiates longer infrared wavelength.<br>
3. Longer wavelength is absorbed by the greenhouses gases and they trap heat energy and reradiate back to the Earth, warming the surface<br>

<br>
→Without greenhouse effect, Earth would be at -63*C instead of 14*C.<br>
→It is the enhanced greenhouse effect that is causing the global warming.
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<tr><td>5.18</td><td>understand how knowledge of the carbon cycle can be applied to methods to reduce atmospheric levels of carbon dioxide</td></tr>
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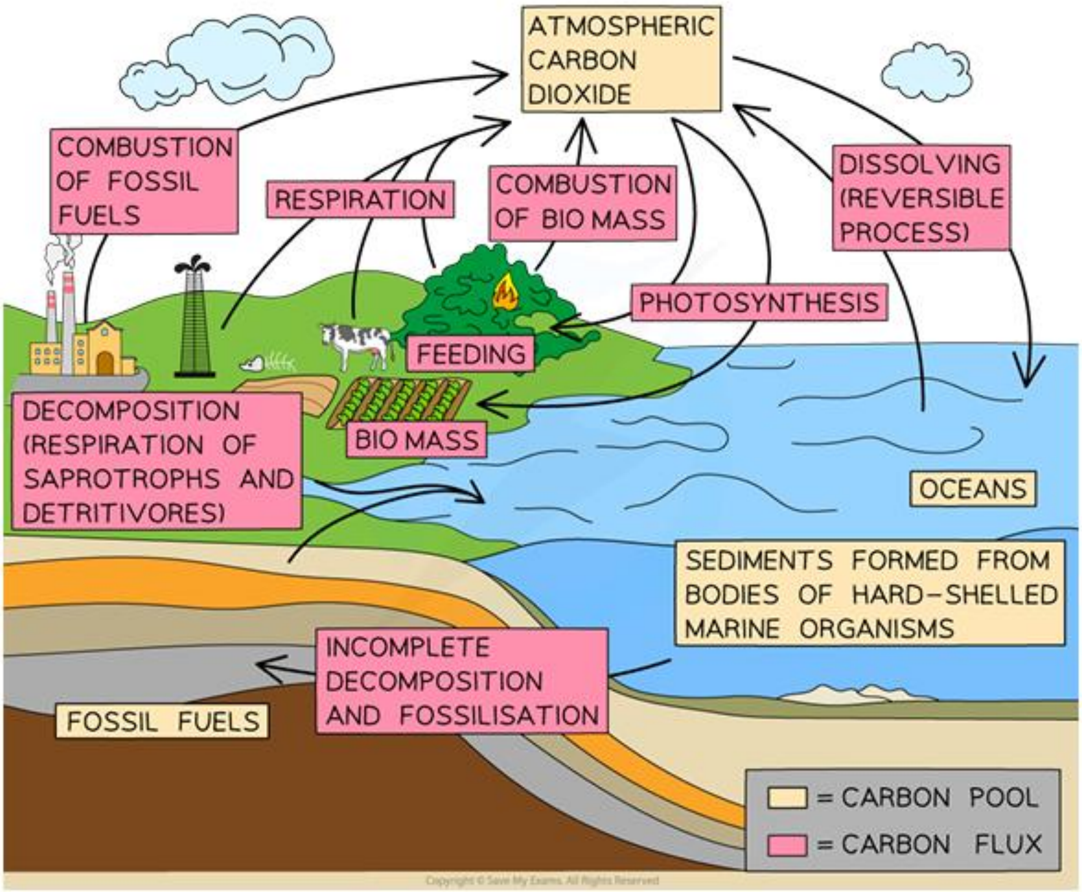


**Storage of Carbon:**

**As Carbon Sink:** a reservoir where carbon dioxide is removed from atmosphere and locked into organic and inorganic compounds
→ Humus, Rocks, Fossil Fuels, Inside the organism body as biomass, Oceans, Coral reefs, Limestone.

**Absorption of Carbon:**

**Photosynthesis:** Light-independent reaction fixes carbon dioxide into GP

**Emission of Carbon:**

- **Aerobic Respiration**
- **Decomposition**
  → Break down organic matter with digestive enzymes
- **Combustion of Fossil Fuels**
- **Volcanic Eruptions**
- **Weathering**
- **Forest Fires**

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| **5.19** | (i) understand that data can be extrapolated to make predictions and that these are used in models of future climate change<br>(ii) understand that models for climate change have limitations |

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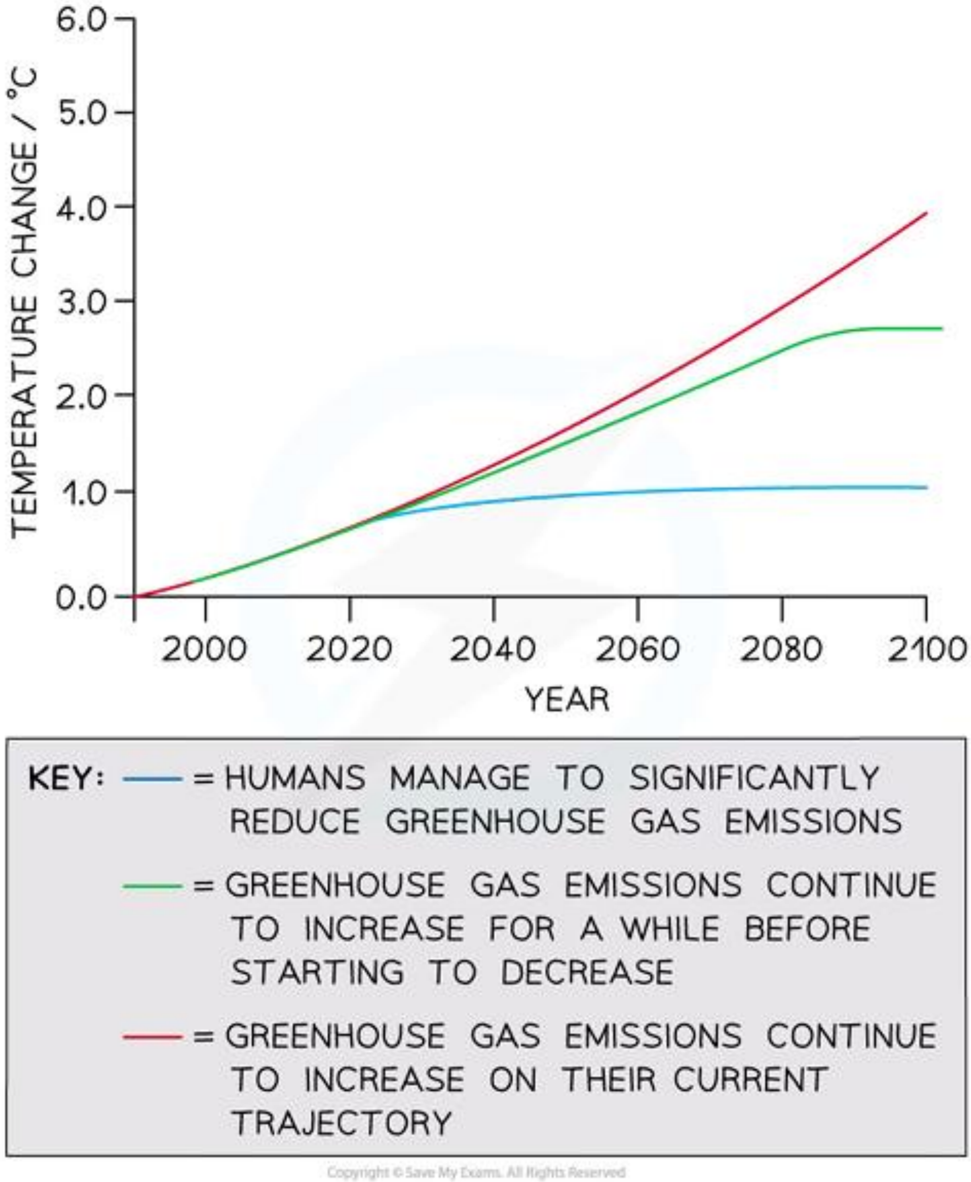
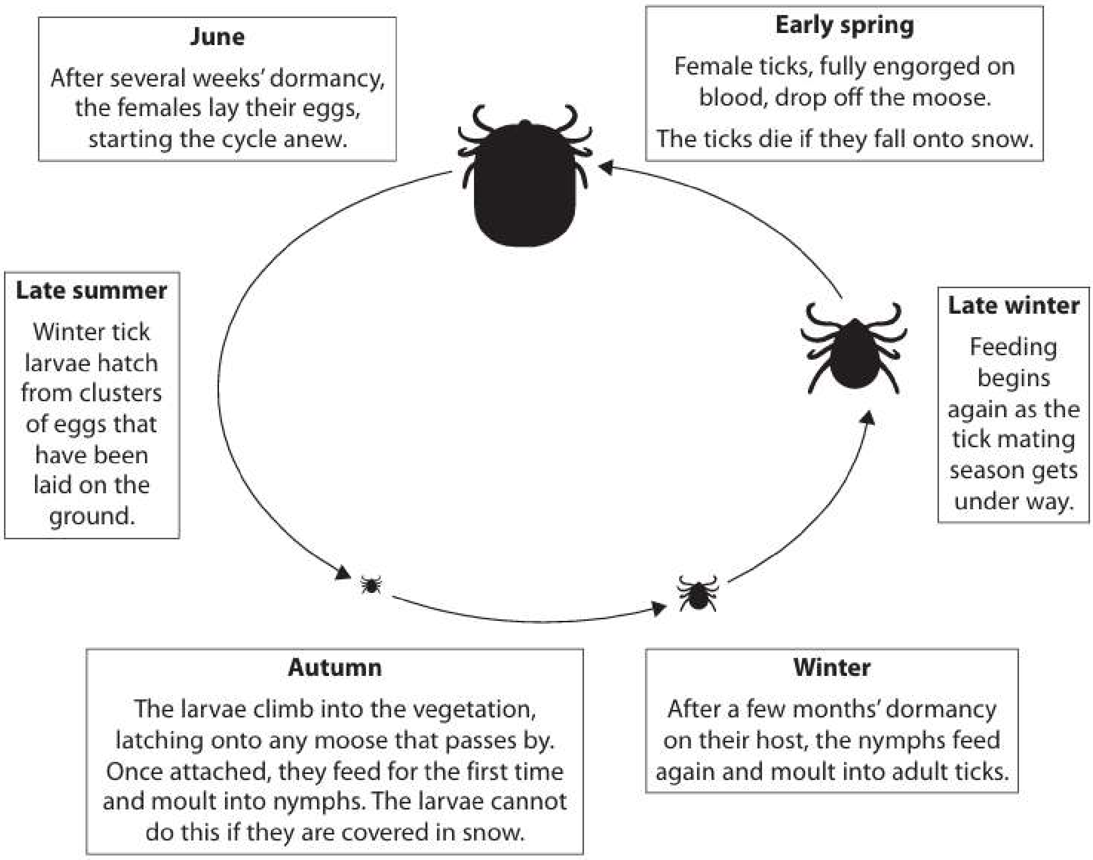
| **i** | **Extrapolation: Use line of best fit** from existing data relating to global warming to make **predictions** about global temperatures<br>→ This allows to build **models** to show how the climate may change in the future<br><br> |
| **ii** | **Limitations of Prediction Models**<br>- Other new factors may affect climate the future<br>- No one knows which scenario is most likely to occur<br>- Future technologies might be successful at removing greenhouse gases from the atmosphere |
| **5.20** | understand the effects of climate change (changing rainfall patterns and changes in seasonal cycles) on plants and animals (distribution of species, development and lifecycles) |
| | **Effect of Climate Change on Plants and Animals**<br>1. Change rainfall patterns: Some areas rain a lot which might lead to flood which destroys crops and disrupt habitats; Some area might not rain at all which results in drought which also makes unable to grow crops and dehydration can occur<br>2. Changes in seasonal cycles: Organisms are adapted to the timing to the seasons to mate, to hibernate and to prey; Changing |

- ➔ Both affect distribution of species and development of life cycles
- ➔ Both can result in migration of species to less hot and safer areas due to climate change.
- This might be due to the enzymes not working efficiently or their prey migrating to other places.
- This can also avoid dehydration.

*(ii) The diagram shows the life cycle of the tick.




The ticks feed on the blood of the moose. One tick can remove 200 to 600 times its body mass.

The ticks irritate the moose. The moose will scratch against trees causing large clumps of fur to fall off.

Explain how global warming might affect the life cycle of ticks and result in the decline in the number of moose.

Use all the information in the question to support your answer.

(6)

Global Warming:
- global warming will increase the temperature of the earth's {surface / atmosphere}
- winters will get warmer so less snow
- winters will get shorter so snow present for fewer days

Effect of Temperature on ticks:
- warmer conditions decrease life cycle time
- fewer ticks will die in the snow in early spring
- more females to lay eggs
- larvae less likely to be covered in snow in autumn
- so, more larvae become nymphs

<table>
<tr><td></td><td>*Check the mark scheme for the full interpretation: the answers are extracted just for this specification point.</td></tr>
<tr><td>5.21</td><td>understand the effect of temperature on the rate of enzyme activity and its impact on plants, animals and microorganisms, to include $Q_{10}$</td></tr>
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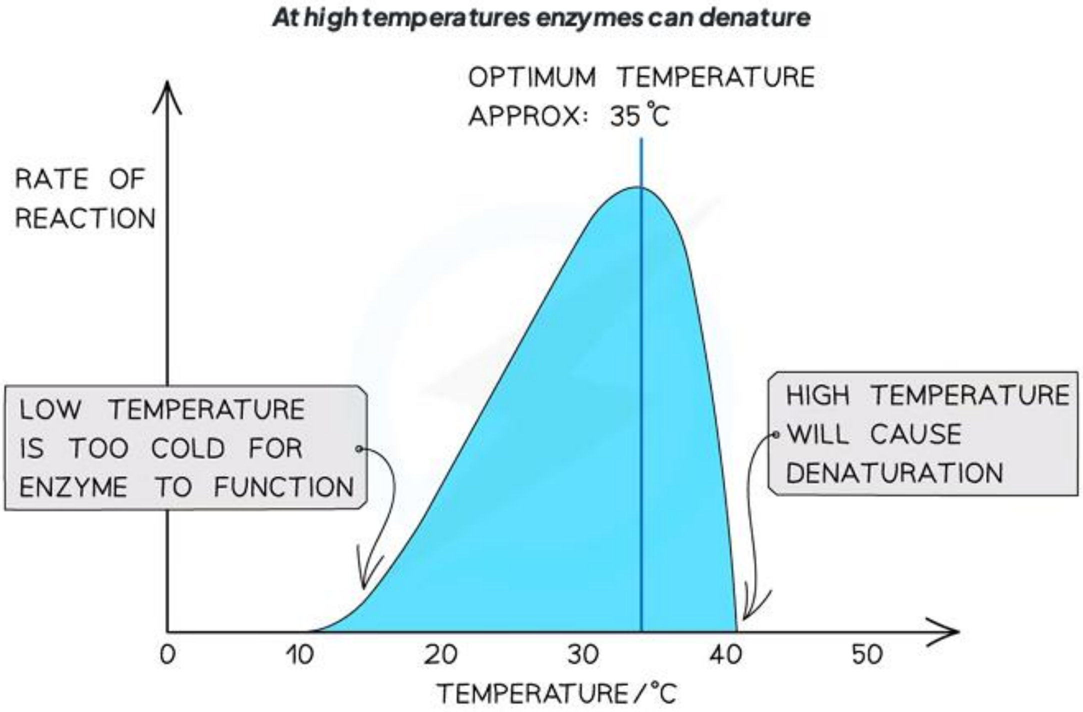


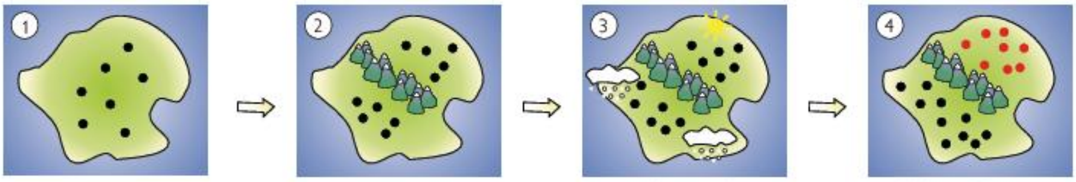

Effect of Temperature on Enzyme:<br>
- As temperature increases, rate of enzyme reaction increases<br>
- This is because more kinetic energy leads to more frequent collisions between enzymes and substrates<br>
- So, DNA synthesis, mitosis, protein synthesis, respiration would be faster, contributing to faster growth<br>
<br>
$Q_{10}$ Value<br>
- Temperature coefficient or $Q_{10}$ value for a reaction shows how much the rate of a reaction changes when the temperature is raised by 10° C<br>
- At temperatures before the optimum, a $Q_{10}$ value of 2 means the rate doubles when the temperature is raised by 10° C.

$$Q_{10} = \frac{\text{rate of enzyme activity at } T + 10^0\text{C}}{\text{rate of enzyme activity at } T^0\text{C}}$$
</td></tr>
<tr><td>5.22</td><td>CORE PRACTICAL 12<br><br>Investigate the effects of temperature on the development of organisms (such as seedling growth rate or brine shrimp hatch rates), considering the ethical use of organisms.</td></tr>
<tr><td>5.23</td><td>understand how evolution (a change in allele frequency) can come about through gene mutation and natural selection</td></tr>
</table>

**Evolutionary Mechanisms:** causes differences in allele frequencies to accumulate

1. **Mutation**: changes in base sequence of the gene
2. **Natural Selection**: the process where individuals with advantageous traits are more likely to survive and reproduce, increasing the frequency of those alleles in a population
3. **Genetic Drift**: random change in allele frequencies in a population due to chance events, especially in small populations such as population bottleneck or Founder's effect

**5.24** understand how isolation reduces gene flow between populations, leading to allopatric or sympatric speciation




Population of individuals ● = individual organism → Physical barriers stop interbreeding and gene flow between populations. → Populations adapt to new environments. → Allele and phenotype frequency change leading to development of new species.

**Species:** Group of living organisms that can reproduce to give fertile offspring
**Hybrid:** Resulting offspring of two different species interbreeding (infertile)
**Subspecies:** Varieties of the same species that can still reproduce to give fertile offspring despite having phenotypic or behavioral differences

**(2) Types of Speciation:**

1. **Allopatric Speciation:** formation of new species due to geographical isolation that prevents gene flow between populations
2. **Sympatric Speciation:** formation of new species within the same geographical area due to reproductive isolation (e.g. ecological, behavioral, or temporal differences)

**Speciation** requires (2) conditions:

1. Isolation Mechanism (Prevents Gene Flow)
2. Evolutionary Mechanism (Change Allele Frequencies)

**1. Isolation Mechanism:** a process that prevent gene flow (exchange of alleles) between populations.

Types of Isolation Mechanism:

1. Geographical Isolation (Allopatric)
2. Ecological Isolation (Sympatric)
3. Seasonal Isolation (Sympatric)
4. Behavioral Isolation (Sympatric)
5. Mechanical and Gametic Isolation (Sympatric)

**2. Evolutionary Mechanisms:** causes differences in allele frequencies to accumulate

1. Mutation
2. Natural Selection
3. Genetic Drift

**Isolation Mechanism + Evolutionary Mechanism → Genetic Differences → Reproductive Isolation → Speciation**

| | |
|---|---|
| **5.25** | understand the way in which scientific conclusions about controversial issues, such as what actions should be taken to reduce climate change, or the degree to which humans are affecting climate change, can sometimes depend on who is reaching the conclusions |
| | **Why some people consider anthropogenic climate change theory is controversial:**<br><br>- Just because there is a correlation between global temperature and increasing greenhouse gas emission, does not mean there is a causation<br>- Certain politicians and business people have different opinions and want to continue their activities<br>- Other activities such as volcanic eruptions and solar flares can cause climate change as well<br>- Due to lack of education and lack of trust in scientists |
| **5.26** | understand how reforestation and the use of sustainable resources, including biofuels, are examples of the effective management of the conflict between human needs and conservation |
| | **Sustainability:** production of products that are carbon neutral and biodegradable so that the future generation do not need to compromise for that<br><br>**How can effects of anthropogenic climate change be reduced:**<br><br>**1.Reforestation:** more plants will absorb more carbon dioxide for photosynthesis and tresses are carbon sinks<br>**2.Biofuel/ Solar/Wind Power:** reduce carbon dioxide released into the atmosphere<br>**3.Reduce cattle being farmed:** reduce the methane being released into the atmosphere<br>**4.Use GM crops:** more drought/insect resistant crops so grow higher yield and prevent starvation<br>**5.Recycle Plant Material as Fertilizers:** prevents eutrophication, bioaccumulation due to artificial fertilizers |

## Notice

- In this booklet, mark scheme points from past paper questions (mostly 2020-2023 and a few from 2010-2019) are included for each specification point to connect the syllabus and the past paper questions.
- For less frequently examined specification points, Pearson Edexcel IAL textbook and Save My Exams (available at savemyexams.com) notes have been the main reference of this booklet.
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